Problem 5

Problem 5

Problem 1 *Consider a square with a side x. Let A be the area of the square.*

(a) *Compute ΔA, the change in area if the side increases by $\Delta x = dx$.*

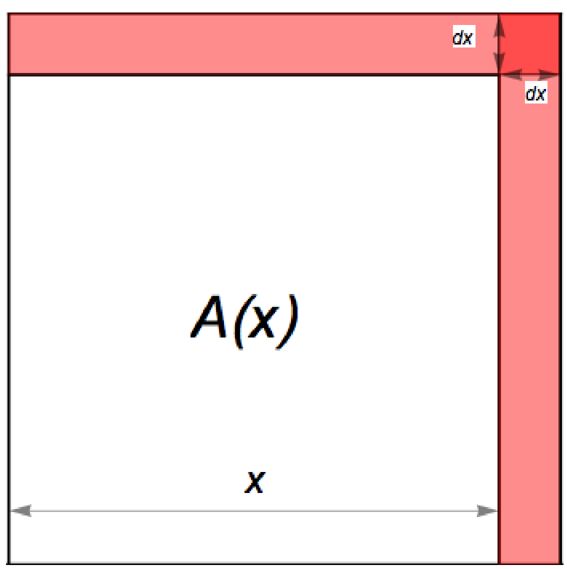
Explanation. $\Delta A = A(x + dx) - A(x) = (x + dx)^2 - x^2 = x^2 + 2x \cdot dx + (dx)^2 - x^2 = 2x \cdot dx + (dx)^2$

(b) *Compute dA, the differential of A at x, and compare it to ΔA.*

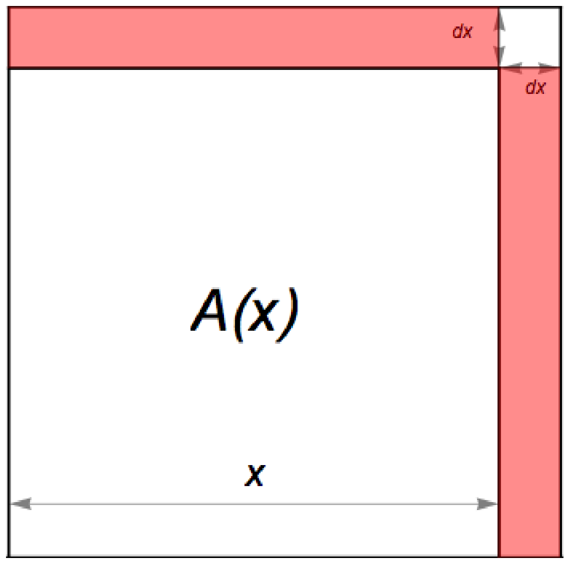
Explanation. $dA = A'(x)\,dx = 2x \cdot dx$

$\Delta A = [2x \cdot dx] + (dx)^2 = dA + (dx)^2$

(c) *In the figure below the shaded part represents the change ΔA. Shade the part that represents dA.*



Explanation. Since $dA = A'(x)\,dx = 2x \cdot dx$, we shade the part as shown in the figure below.



The picture illustrates that $\Delta A \approx dA$.